

Alex Millar

Design Mechanical Engineering Student

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Profile

I am a final-year MEng Integrated Design Engineering student at the University of Bath. My work has been automotive and motorsport throughout: a year at Bentley Motors across Production Planning and the Continental GT assembly line, and FEA on Team Bath Zero's MotoStudent 2027 entry. I design across mechanical, electrical and software interfaces, from Ansys structural analysis and MATLAB numerical modelling to Simulink control architectures and hardware prototyping. A rotational programme suits how I have worked so far, giving me the exposure to develop across each area and to work out where that breadth is most useful.

Education

Master of Engineering in Integrated Design Engineering University of Bath

October 2022 - July 2027

Predicted: First Class Honours.

Originally enrolled in Integrated Mechanical and Electrical Engineering (October 2022 – August 2025)

Relevant units: mechatronics, design optimisation, numerical methods and simulation, reverse engineering, CFD.

Engineering Practice Summer School Shanghai Jiao Tong University and University of Bath

July 2026

Selected for the inaugural cohort. Aeroengine practice covering CFM56 disassembly, compressor assembly and borescope inspection, alongside visits to robotics and hydrogen energy manufacturers.

Summer School Programme Imperial College London

July 2021 - August 2021

Completed an intensive engineering programme focused on applied problem-solving, laboratory work and exposure to university-level engineering concepts.

Lycée Français Charles De Gaulle

September 2015 - June 2022

- 5 A-levels: French (A*), Spanish (A), Maths (A), Physics (B), History (B)
- French Native Speaker Exam: A*

Core Competencies

- Strategic Thinking
- Leadership & Ownership
- Stakeholder Management
- Adaptability & Resilience
- Structured Problem Solving
- Continuous Improvement Mindset
- Teamwork and Communication

Technical Skills

- Inventor, AutoCAD
- Ansys Workbench (FEA and CFD)
- C++, Python, MATLAB
- 3D Printing, Prototyping
- Excel, Powerpoint, Presenting
- MATLAB, Simulink
- Mechanical & Electrical Integration

Languages

- English (Native) - French (Native) - Spanish (Fluent) - German (Basic)

Experience

Team Bath Zero - FEA Engineer, MotoStudent 2027

September 2025 - Present

- Run static structural analysis of the motorcycle wheel in Ansys Workbench, importing Inventor geometry via STEP and splitting contact-patch faces to apply realistic loading.

- Derived the design load case from first principles, applying 0.30 MPa across a 7489 mm² contact patch to represent a 230 kg bike-and-rider mass.
- Resolved mesh and solver issues including rigid body motion, body fragmentation and node-count limits, producing analyses the design team can act on within licence constraints.

Bentley Motors - Production Planning, Business and Strategic Planning

August 2024 - August 2025

- Coordinated production planning responses to strategic and engineering changes across Bentley and the VW Group, ensuring accurate cross-departmental input and timely submissions, and built the documentation systems supporting material approval and process traceability.
- Worked directly on the Continental GT assembly line, supporting wheel, seat, FEM and door fitting, as well as quality checks and road-test preparation. Identified a source of wastage and worked to put in place cost saving measures.
- Supported the Physical Mock-Up team with prototype build activities and cross-functional coordination to assess design and manufacturing feasibility.

The King's Trust - Event Organiser, Bentley Motors Visit

March 2025 - July 2025

- Led end-to-end planning of a full-day visit for 10 students, coordinating the Wood, Trim, Paint and Final Assembly departments to deliver hands-on workshops, factory tours and demonstrations.
- Ran a CV and cover-letter masterclass alongside Bentley early-careers specialists, presenting to the group and managing stakeholder communication and on-site flow across the day.

Team Bath Hydrobotics - Operations Team Leader

December 2024 - August 2025

- Implemented a new change-management system and coordinated planning for the team's attendance at an international competition.

ENGin - Mentor

October 2023 - January 2026

- Delivered weekly English-speaking and career support to Ukrainian learners.

Projects

Tilt-Wing Drone Design Project - Project Leader - Control System Lead

MATLAB, Simulink, Inventor, Lucid

- Led a group design and business project with MBDA Missile Systems, defining system requirements and implementing the V-Model systems engineering approach.
- Designed control functions and subsystem behaviours using MATLAB and Simulink modelling.

Transmission Shaft Design

Mechanical Design, Stress Analysis, Material Selection, CAD

- Sized synchronous belt drives and pulley ratios to deliver 150 RPM at the pump within centre-distance constraints, and specified deep-groove ball bearings from dynamic load ratings and lifetime calculations.
- Conducted combined bending and torsion analysis to the Von Mises criterion with stress-concentration factors for shoulders, keyways and circlip grooves, selecting EN16 low-alloy steel to balance strength, machinability and cost.

Gantry Laser Targeting System

Mechatronics, Computer Vision, Simulink–Python Integration

- Integrated Python ArUco detection with Simulink motor control over TCP, enabling real-time target identification, distance estimation and coordinated laser firing.
- Built the full control architecture including end-switch homing, PWM stepper actuation, Stateflow logic and a custom MOSFET laser PCB, achieving reliable scanning across a 42.5 cm range.

Awards

- Bentley BeUnited Award Winner
- W.O. Bentley Industrial Placement of The Year Shortlist
- The Bath Award
- Outstanding Engagement Award - Imperial College
- Duke of Edinburgh Silver/Bronze Award